Constrained optical matching and error sensitivity of a booster-to-storage-ring transfer line

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Working draft for Journal of the Korean Physical Society. This version reports archived simulation results. Italic revision notes identify analyses and author inputs still needed; they are not completed findings.

Abstract

Optical matching in a booster-to-storage-ring transfer line must deliver a specified phase-space distribution while controlling the beam envelope. We examine a 4 GeV transfer-line model with nine independently adjustable quadrupoles, using normalized optical residuals and bounded numerical searches followed by physical feasibility checks. For the archived exit target, the selected solution matches both transverse planes to floating-point precision, with peak horizontal and vertical beta functions of 31.44 m and 50.75 m, respectively, below the imposed 60 m ceiling. Four fixed-optics error ensembles containing 550,000 realizations give a median horizontal mismatch of 0.002086, with a 95% interval of 0.002074–0.002099. Five realizations violate the implemented optical feasibility criteria, corresponding to a probability estimate of 9.09×10^{-6} with an exact 95% interval of 2.95×10^{-6}–2.12×10^{-5}. The historical sensitivity ranking places input beta errors ahead of quadrupole gradient and energy errors under the assumed budget. These results characterize an implemented optics model rather than physical beam loss or storage-ring capture. They establish a matched reference and quantify statistical precision, while identifying the boundary, error-coverage and transport validation needed to turn the model into practical operating tolerances.

Keywords: transfer-line optics; quadrupole matching; beam transport; error sensitivity; accelerator simulation

1 Introduction

Low-emittance storage rings require careful control of the beam distribution presented to their injection region. The design of diffraction-limited rings involves strong focusing and limited nonlinear acceptance [1]. The Korea Light Source design provides a relevant 4 GeV setting for this problem [2]; booster injection and ramping have also been studied for the Korea-4GSR project [3]. A booster-to-storage-ring (BTS) line must transport the extracted beam to a declared boundary with suitable transverse optics and dispersion. Its performance depends on both the nominal match and the response to errors in the incoming beam and magnets.

A small exit mismatch does not alone establish a useful transfer-line design. Large beta functions elsewhere in the line can reduce aperture margin, while sensitive quadrupole combinations

can make a nominal solution difficult to maintain. Moreover, the significance of any reported tolerance depends on which errors actually act in the evaluator and which observables define failure. Matching, transmission through the BTS and capture in the storage ring are separate quantities.

This draft examines the deterministic matching and fixed-optics tolerance campaign archived in the nkm-injection repository. The central question is whether its nine-quadrupole model reaches a specified exit target within the configured limits, and what the existing error ensemble establishes about that solution. We report the selected configuration, its statistical uncertainty and the limits of the implemented model. Nonlinear-kicker field effects, stored-beam transparency and multiturn storage-ring capture belong to the companion NKM study and are outside the present results.

Draft revision: B01: expand the comparison with published BTS matching and correction studies, including relevant JKPS work. The additional physical contribution should be a validated margin comparison and interpretable correction analysis; use of established optimization algorithms alone is not a novelty claim.

2 Transfer-line model and boundary conditions

2.1 Lattice and coordinates

The model is constructed with Accelerator Toolbox and its Python interface, pyAT [4, 5]. Its nominal energy is 4 GeV and its length is 21.789 m. Nine 0.2 m quadrupoles are arranged in three triplets, named q11–q13, q21–q23 and q31–q33. The sequence also contains an extraction kicker, two septum elements and three bending magnets. The nominal bending angles in sequence are +7.500, +88.500, −111.701, +176.000, −111.701 and +88.500 mrad; magnet lengths and edge angles are part of the archived configuration.

Positions are expressed in metres and normalized quadrupole strengths $K = G/(B\rho)$ in m^{-2}, where G is the magnetic gradient in T/m and $B\rho$ is the magnetic rigidity in T m. Particle tracking uses AT coordinates $(x, p_x, y, p_y, \delta, ct)$, where p_x and p_y are normalized canonical momenta and δ is the relative momentum deviation. Linear transverse optical quantities use the corresponding paraxial convention. The dispersion derivative D'_x is dimensionless.

The line configuration places square aperture elements with half-widths 19.35 mm at its ends and 30 mm at two interior locations. These are model apertures; continuous vacuum-chamber geometry and measured clearances have not been established for the present article. The preliminary MAD-X file has ten quadrupoles and cannot serve as an independent equivalent-lattice validation without reconciling its sequence.

2.2 Entrance and exit optics

Table 1 gives the boundary values used in the archived matching configuration. The target is treated here as a declared BTS-exit target. It must not be identified with the storage-ring optics at the NKM centre, since those values and the intervening transport differ.

The historical constraint configuration assumes horizontal and vertical emittances of 100 and 10 nm rad and a relative energy spread of 0.0011. These are sensitivity-study inputs. Later author documentation identifies a 7.67 nm rad horizontal design booster emittance; the archived ensembles are not relabelled as simulations of that beam. A design-beam transmission study

Table 1: Archived entrance optics and declared exit target. Beta and dispersion are in m; alpha and dispersion derivative are dimensionless. The physical provenance of the target boundary remains to be confirmed.

Quantity	Entrance	Exit target
β_x [m]	7.560000	2.336495
α_x [1]	1.523100	−0.016335
β_y [m]	12.269000	4.256241
α_y [1]	−1.654700	0.017772
D_x [m]	0.276200	0.080868
D'_x [1]	−0.065700	0.047472

requires a consistent specification of emittance, coupling, momentum spread, centroid, distribution and extraction optics.

Draft revision: B02–B04: confirm the source and observation point of the boundary optics, reconcile AT and MAD-X geometry, provide a sourced layout figure and validate transport independently. Establish the main beam specification and physical chamber before reporting design-beam transmission.

3 Matching and uncertainty methods

3.1 Normalized matching and feasibility selection

For a quadrupole vector $\mathbf{K}$ and exit observables $\mathbf{O} = (\beta_x, \beta_y, \alpha_x, \alpha_y, D_x, D'_x)$, the implemented objective is

$$r_i(\mathbf{K}) = \frac{O_i(\mathbf{K}) - O_{i,\mathrm{target}}}{s_i}, \qquad J(\mathbf{K}) = \sum_{i=1}^{6} r_i^2. \tag{1}$$

The normalization scales are 0.05 m for each beta residual, 0.01 for each alpha residual, 0.002 m for dispersion and 0.001 for its derivative. These make J dimensionless; they are numerical scales rather than demonstrated operating tolerances.

The archived workflow first uses bounded least squares. When its result is successful and away from the strength bounds, SLSQP refines the same scalar merit. The configured budget is 100 for the least-squares maximum function evaluations and SLSQP maximum iterations. The current SLSQP call supplies strength bounds but no nonlinear constraint functions. Other physical criteria are evaluated separately, and the final selection prioritizes feasible candidates. This distinction matters when comparing the procedure with a fully constrained search.

The outer random seed is 42. The first of three starts uses nominal settings; two additional starts are sampled within the family bounds. q11 and q33 have bounds $[-2.5, 2.5]$ m^{-2}, while the other families use $[-3, 3]$ m^{-2}. The archived constraint configuration also specifies a 60 m beta ceiling, 1.5 m dispersion ceiling, a 1.2 T pole-tip field limit and beam-envelope and injection-geometry criteria. Their interpretation as hardware specifications requires source confirmation. Successful optimizer termination and physical feasibility are recorded separately.

For uncoupled Twiss comparison, let

$$T_u = \begin{pmatrix} \beta_u & -\alpha_u \\ -\alpha_u & (1+\alpha_u^2)/\beta_u \end{pmatrix}, \qquad \mathcal{M}_u = \tfrac{1}{2}\,\mathrm{Tr}(T_{u,\mathrm{target}}^{-1} T_{u,\mathrm{out}}) - 1, \tag{2}$$

where $u = x, y$. The matrices are normalized to equal emittance; $\mathcal{M}_u$ is dimensionless and vanishes for a matched plane.

3.2 Archived tolerance ensemble

Three independent runs use 50,000 realizations each with seeds 42, 123 and 777; a follow-up uses 400,000 realizations with seed 2026. All select the same optimization and error-budget hashes. The quadrupole settings remain fixed while errors are sampled. The analysis contains no capture callback and no particle-through-aperture transmission measurement. Its optical failure criterion is

$$\max(\beta_{x,\max}, \beta_{y,\max}) > 60.01 \text{ m} \quad \text{or} \quad \mathcal{M}_x + \mathcal{M}_y > 2.05, \tag{3}$$

including the stored 0.01 m beta and 0.05 mismatch allowances. This definition is less extensive than the nominal design's full feasibility checks.

Quantile intervals are calculated from order statistics using a binomial coverage construction. Median bootstrap checks use 20,000 replicates, with resampling seeds 1729, 2718 and 31415. For k failures in N independent realizations, the two-sided exact 95% interval is obtained from beta-distribution quantiles at 0.025 and 0.975 with parameters $(k, N-k+1)$ and $(k+1, N-k)$, respectively, with endpoint conventions for $k = 0$ or $k = N$. Wilson score intervals provide a cross-check [6]. These intervals describe sampling uncertainty conditional on the chosen model and budget, not uncertainty in the physical assumptions.

3.3 Error coverage

The historical budget includes relative quadrupole-gradient errors with standard deviation 10^{-3}, relative energy errors of 10^{-3}, input beta errors of 5%, quadrupole offsets of 100 μm and rolls of 0.5 mrad. The coverage audit reports effects on the optics from gradient, energy, beta and roll terms, with an offset contribution to peak beta through the implemented approximation. Dipole-strength errors, input centroid jitter and longitudinal placement errors do not affect the reported optical metrics in that evaluator. The same driver also computes an NKM stored-kick diagnostic; this diagnostic is excluded from the BTS results.

Consequently, an inactive error cannot be assigned a generous physical tolerance from a zero numerical response. Coupled transport and orbit-dependent clearance need explicit validation, particularly for roll and alignment errors.

Draft revision: B07: generate a machine-readable perturbation coverage audit and implement the missing physical responses needed for BTS transport claims. Document distributions, correlations and sources for each error amplitude.

4 Matched design and archived performance

4.1 Selected quadrupole settings

The archived search finds one feasible candidate among three starts, with the termination message "Optimization terminated successfully." Its final merit is 1.2906×10^{-13}, and both mismatch values print as zero at floating-point precision. The horizontal dispersion residual is 4.382×10^{-11} m. This is a numerical match to the declared target, not evidence of a global optimum or an experimentally realized setting. Table 2 reports the selected strengths.

Table 2: Nominal and selected quadrupole strengths, and family bounds, all in m^{-2}. Values are exported from the selected archived summary and configuration. Rounded settings have not been independently revalidated here.

Family	Nominal K	Selected K	Bounds
q11	+0.448572	-0.367525	[-2.5, 2.5]
q12	-1.026778	-1.216433	[-3.0, 3.0]
q13	+0.887640	+1.586072	[-3.0, 3.0]
q21	-1.066465	-1.068373	[-3.0, 3.0]
q22	+1.488384	+2.823108	[-3.0, 3.0]
q23	-0.669894	-2.490968	[-3.0, 3.0]
q31	+0.589886	+2.042678	[-3.0, 3.0]
q32	-1.168702	-1.434271	[-3.0, 3.0]
q33	+0.941655	+0.351115	[-2.5, 2.5]

The selected peak beta functions are 31.4368 m horizontally and 50.7459 m vertically. Relative to the configured 60 m ceiling, the vertical plane has the smaller beta margin. That margin is not itself a physical clearance: for an uncoupled beam, the rms horizontal size obeys

$$\sigma_x^2(s) = \beta_x(s)\varepsilon_x + D_x^2(s)\sigma_\delta^2. \tag{4}$$

A chamber check must also include the centroid orbit, vertical or coupled covariance, a declared beam-coverage factor and the aperture shape.

Draft revision: B05 and B08: recompute baseline and matched optics from the same frozen configuration; compare a design chosen for envelope or error margin using equal physical assumptions. Add optics/dispersion and clearance profiles. No new robust design or nominal transmission gain has yet been established.

4.2 Tolerance distributions

All 550,000 requested evaluations in the selected historical ensemble are numerically valid. Table 3 gives pooled mismatch and peak-beta quantiles, omitting the stored-kick observable. The median horizontal mismatch is 0.00208629, with a 95% interval $[0.00207403, 0.00209890]$. Its upper tail is appreciably larger than its median, so a single central estimate does not describe every realization.

Table 3: Pooled fixed-optics results from 550,000 archived realizations. The interval is the 95% order-statistic interval for the median. Mismatch is dimensionless. These metrics do not measure particle transmission.

Observable	Median	Median interval	p95	p99
$\mathcal{M}_x$ [1]	0.002086	[0.002074, 0.002099]	0.016482	0.028814
$\mathcal{M}_y$ [1]	0.002499	[0.002483, 0.002514]	0.020701	0.035932
$\beta_{x,\max}$ [m]	31.438418	[31.430758, 31.445751]	35.564889	37.521376
$\beta_{y,\max}$ [m]	50.751986	[50.746458, 50.757604]	53.831455	55.275722

Five realizations fail the optical criteria in Eq. (3), giving $\hat{p} = 5/550000 = 9.09 \times 10^{-6}$ and an exact 95% interval of $[2.95 \times 10^{-6}, 2.12 \times 10^{-5}]$. The small count leaves considerable relative uncertainty in this probability. It must not be interpreted as five lost bunches, an accelerator reliability specification, or a measurement of ring injection efficiency.

5 Error sensitivity and statistical precision

5.1 Independent seeds and prefix checks

Table 4 records the independent runs and their predeclared absolute median-stability target of 10^{-5}. The 50,000-sample runs compare prefixes of 25,000 and 50,000 samples; the follow-up compares 200,000 and 400,000. All three smaller runs miss the target. The follow-up meets it with an absolute change of 2.395×10^{-6}.

Table 4: Per-run sample counts, optical failure counts, median horizontal mismatch and prefix changes. "Met" refers only to the 10^{-5} absolute median-stability target, not to convergence of rare failures or tail quantiles.

Seed	Samples	Failures	Median $\mathcal{M}_x$	Prefix change	Target
42	50,000	0	0.00210273	20.830×10^{-6}	Not met
123	50,000	1	0.00207934	11.436×10^{-6}	Not met
777	50,000	0	0.00210260	10.567×10^{-6}	Not met
2026	400,000	4	0.00208351	2.395×10^{-6}	Met

A passing median-prefix check does not certify every statistic. The pooled median interval has a half-width of about 1.24×10^{-5}, while the failure-probability interval remains broad relative to its estimate. Sampling precision must therefore be assessed separately for central mismatch, extreme envelopes and rare feasibility failures. Increasing bootstrap count cannot replace independent physical realizations or correct an incomplete error model.

5.2 Sensitivity interpretation and correction analysis

Under the historical budget, the one-at-a-time ranking places input beta mismatch first, followed by quadrupole-gradient and energy errors. This ordering is conditional on the specified error amplitudes and the scalar merit. It does not rank errors absent from the optical evaluator, nor does it determine an optimal hardware specification independently of beam and aperture assumptions.

An interpretable correction study should use a dimensionless response matrix,

$$A_{ij} = \frac{\Delta K_{j,\mathrm{ref}}}{s_i} \frac{\partial O_i}{\partial K_j}, \qquad A = U \Sigma V^{\mathsf{T}}, \tag{5}$$

where s_i are declared optical scales and $\Delta K_{j,\mathrm{ref}}$ are meaningful strength-change scales. Its singular vectors can identify correction combinations and weakly observable freedoms, but finite-step reevaluation and constraint checks are needed before treating them as usable knobs. The raw mixed-unit Jacobian condition number is not used as a tolerance result here.

Draft revision: B06, B09 and B10: validate finite-difference steps and singular rank, measure finite correction range, scan relevant error amplitudes, and compare uncorrected/corrected transport on held-out static errors with independent shot jitter. Add sensitivity and correction figures only from completed analyses.

6 Discussion

The archived solution demonstrates that the chosen linear BTS model can meet its declared optical target with peak beta below the configured ceiling. The error ensemble provides a reproducible

conditional characterization of mismatch and envelope fluctuations. Its principal practical message is that input optics must be controlled alongside magnet settings; a nominally exact match does not remove sensitivity to the extracted beam.

Several limits prevent direct translation to commissioning requirements. The target boundary and lattice provenance need source confirmation; the historical beam emittance differs from the later design specification; and some sampled errors have no effect on the reported optical quantities. The nominal feasibility validator and the ensemble optical-failure rule also test different sets of conditions. Physical transmission requires explicit particles and apertures, including orbit and coupling effects, rather than a beta threshold alone.

The archived multi-objective study is not used to support the present results. Its dispersion objective combines D_x in metres with dimensionless D'_x, and its legacy finalist procedure does not establish physical BTS transmission. A future trade-off study should normalize both residuals consistently and compare physically validated designs under common constraints and held-out errors. It need not use a genetic algorithm if deterministic searches expose the relevant margin trade-off.

Draft revision: B08–B10: replace these prospective points with a quantified physical design and correction comparison before claiming the planned article's additional contribution. B01–B12 remain pending; this first draft does not complete them.

7 Conclusions

For the archived 4 GeV nine-quadrupole BTS model, the selected bounded-search solution matches the declared exit optics to floating-point precision and has peak beta functions of 31.44 and 50.75 m. Under the implemented fixed-optics error budget, 550,000 realizations yield a median horizontal mismatch of 0.002086 and five optical feasibility failures. The larger follow-up meets the specified median-prefix target, while rare-failure uncertainty remains distinct. These findings establish a reference for further transport and correction studies. They do not establish physical beam transmission, hardware tolerances or storage-ring capture. A source-confirmed boundary model, active-error audit and held-out margin/correction comparison are needed to substantiate the final BTS article's engineering conclusions.

Statements and declarations

Author contributions. *Draft revision: Author to confirm contributions and final authorship.*

Funding and competing interests. *Draft revision: Author to supply funding/grant information and confirm the competing-interest declaration.*

Acknowledgments and generative assistance. This working draft was prepared with OpenAI Codex assistance. Scientific claims are limited to the selected archived artifacts; the author must verify the text and approve the final assistance disclosure and acknowledgments.

Data and code availability. The model and analysis code are maintained in the nkm-injection repository (`https://github.com/cspark7701/nkm-injection`). The selected simulation outputs are ignored by version control and are not supplied by the repository alone. A local

provenance record accompanies this draft. No permanent public archive or BTS referee attachment is claimed to exist.

Draft revision: B12: provide the selected raw-output attachment, source/environment snapshot and independently verified rebuild instructions before submission.

A Draft evidence selection and unfinished assets

The nominal solution is selected from `backups_20261009/production_statistical_01/optimization/`; its summary and adjacent configuration match the hashes recorded in the outline. The tolerance selection is `results/journal_campaign_01/followup_analysis/tolerance_replicates_with_400k.json`. All four raw-summary hashes were checked when preparing the publication values. The exported tables are installed in `generated/archived_results.tex`; `draft_provenance.json` records the source paths, digests and source commit. The current PDF can be compiled from these installed publication files without access to the ignored raw ensembles, but such compilation does not rerun or independently validate their physics.

The six planned main figures are not inserted as simulated or measured results. The layout, optics/envelope profiles, normalized correction responses, margin trade-off, active-error scans and held-out correction/distribution plots remain assigned to B02–B10 in `002_manuscript_tasks.md`. The bibliography below contains six entries selected from the existing audited reference list; the closest BTS literature comparison remains B01 work. This generic one-column article format is for drafting; final JKPS formatting is a B11 deliverable.

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